

ATHANG GHAG

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EXPERIENCE

Retail Insights Executive - Insights & Data Analyst

London, UK

Argos, J Sainsbury's plc

May 2026 – Current

- Analysed and validated large-scale customer and operational datasets across 667 retail stores using SQL in Snowflake and applied data quality checks in Excel to support operational decision-making.
- Conducted A/B testing and statistical analysis to identify trends, investigate operational issues and evaluate in-store activations, contributing to a 1% year-on-year improvement in customer satisfaction scores.
- Automated recurring reporting using Excel Macros and Copilot, investigated data quality issues, and transformed complex, unstructured data into accurate, practical insights for stakeholders and retail field leaders.

Operational Support Specialist - Reporting & Data Analyst

London, UK

Argos, J Sainsbury's plc

April 2025 – May 2026

- Reduced stock movement errors by 4.5% year-on-year by analysing distribution centre data, investigating discrepancies and identifying key loss drivers using hypothesis testing and KPI modelling.
- Identified at-risk stores and fulfilment centres across approximately 10 million records using clustering and time-series analysis in Python to drive operational decisions and reduce disruption during peak retail periods.
- Collaborated with operational and governance stakeholders to deliver an improved data-driven operating model for retail stores, reducing labour costs and increasing store KPI attainment by 3%.

SKILLS

- **Programming:** Python (matplotlib, pandas, numpy, scipy, scikit-learn, keras), SQL(joins, CTEs, aggregations), R, C.
- **Tools & Platforms:** MS Excel (VBA, Macros, PowerQuery), Power BI (DAX, Data Modelling, Information Design), Jupyter Notebook, Snowflake, MongoDB, RStudio, AI (MS Copilot Studio).
- **Data Analysis & Modelling:** KPI Modelling & Analysis, CRISP-DM, ETL Implementation, Customer Insights Analysis, Statistical Modelling & Computing, Predictive Modelling, Multi-Level Regression, Monte Carlo Simulation, Trend Analysis.
- **Data Visualisation & Presentation:** Proficient in creating interactive dashboards and visualisations using Power BI and Excel, with expertise in KPI reporting, performance analysis, and data storytelling. Experienced in delivering stakeholder presentations, documenting technical insights in LaTeX.

EDUCATION

University of Leeds

Leeds, UK

MSc Data Science and Analytics

- High Merit (2.1).
- Key Modules: Statistical Computing, Mixed Models with Medical Applications, Deep Learning (Convolutional Neural Networks, Recurrent Neural Networks), Time Series and Spectral Analysis.
- Applied statistical modelling, regression, time-series analysis, computational methods and machine learning techniques to real-world datasets using Python and R.

K. J. Somaiya College of Engineering, University of Mumbai

Mumbai, India

BEng Electronics and Telecommunication Engineering

- Grade: 8.4/10.0.
- Key Modules: Data Analysis and Interpretation, Digital Signal Processing, Business Analytics, Cloud Computing (AWS, Google Cloud), Big Data Analytics (Hadoop, NoSQL).
- Completed coursework in Data Analysis and Interpretation, Business Analytics and Big Data Analytics, applying analytical methods to academic projects and case studies.

TECHNICAL PROJECTS

Stochastic Differential Equations for Wind Speed Modelling

MSc Dissertation | MET Office

University of Leeds

Partnered with the MET Office to model and forecast the uncertain behaviour of wind speed.

- Designed a predictive model using an Ornstein–Uhlenbeck stochastic differential equation and the Milstein method, improving wind speed forecasting accuracy by 15%.
- Developed Python-based analytical pipelines to collect, validate and process meteorological datasets, fitting annual wind speed trends to a generalised gamma distribution.
- Implemented Monte Carlo simulation techniques using NumPy to forecast wind speed and evaluated and validated the predictive model against real-world data for the MET Office.

Speech Emotion Recognition

BEng Capstone Project

K J Somaiya College of Engineering

Coordinated with peers to build a sophisticated model for classifying human emotions using speech.

- Built and evaluated a Convolutional Neural Network to classify 8 human emotions using acoustic features extracted from 5252 audio files.
- Engineered and evaluated combinations of MFCC, Mel and Chroma features, achieving 84.9% accuracy and comparing model performance against findings from relevant research.
- Performed exploratory analysis and visualisation of large-scale audio feature data to identify the most informative acoustic indicators for emotion classification.

LEADERSHIP / EXTRACURRICULAR

Treasurer

Leeds, UK

Residence Life Social Committee, University of Leeds

- Managed a fiscal budget of £10,580, ensuring funds availability for 10 events organised by the residence life social committee.
- Coordinated with a team of 8 members and residents, achieving a 20% average increase in resident participation throughout the academic year.

Team Leader

Mumbai, India

Case Study Competition, ZS Associates

- Accomplished a place in the top 100 teams across the country in an analytical case study challenge organised by the consulting firm.

CERTIFICATIONS

- Hands-On Essentials: Data Warehousing Workshop by Snowflake.
- The Data Scientist's Toolbox by Johns Hopkins University.
- Machine Learning Algorithms: Supervised Learning Tip to Tail by Alberta Machine Intelligence Institute.
- Python for Data Science, AI & Development by IBM.

PROFESSIONAL MEMBERSHIPS

- Royal Statistical Society
- Alan Turing Institute
- Leeds Institute of Data Analytics